

DEV KUMAR

+1-312-532-8223 | devkumar.dklv@gmail.com | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Software Engineer (Java Server / Distributed Systems) with 8+ years of professional server-side Java development building high-performance, scalable, and extensible backend services, including email and messaging delivery platforms and Apache Cassandra operated in production at scale. Privacy and security first mentality with hands-on depth in mTLS, authentication, authorization, and security auditing for modern microservices. Expert in asynchronous programming, non-blocking I/O, multi-threading, NoSQL (Cassandra, Redis), and Docker/Kubernetes on cloud infrastructure, with a strong sense of ownership across the full lifecycle of multiple services from architecture and design to development, testing, and maintenance.

KEY SKILLS

Server-Side Java: Core Java (8, 11, 17), Spring Boot, Spring MVC, Spring Cloud, Hibernate/JPA, Maven, Gradle, asynchronous programming, non-blocking I/O, multi-threading, Java concurrency (CompletableFuture, ExecutorService), REST APIs, clean, efficient, and thoroughly tested code

Security & Privacy: mTLS (certificate authorities, trust stores, certificate rotation), authentication, authorization, security auditing, SIEM (Splunk, Azure Monitor), privacy and security first, RBAC, OAuth 2.0, workload identities, service mesh, encryption in transit and at rest

Distributed Systems: Distributed systems, global services, microservices, event-driven architecture, Kafka, message queues, dead-letter queues, retries with exponential backoff and jitter, idempotency, rate limiting, high-performance and extensible design

NoSQL & Caching: Apache Cassandra (CQL, partition/clustering keys, time-bucketed partitions, replication strategies, consistency levels, compaction, TTL retention, tombstone management, repairs, multi-datacenter replication), Redis, MongoDB, PostgreSQL, caching, hot-partition mitigation

Cloud & Containerization: Docker, Kubernetes, EKS, Helm, ingress gateways, private endpoints, cloud infrastructure, AWS (EC2, S3, IAM, Lambda, CloudWatch), Terraform, rolling deployments, autoscaling

Email & Messaging: SMTP, MIME, SPF, DKIM, DMARC, sender identity, domain verification, delivery orchestration, bounce and complaint processing, suppression lists, provider webhooks, deliverability monitoring

Testing & Ownership: JUnit 5, Mockito, integration testing, CI/CD (Jenkins, AWS CodePipeline), observability, full lifecycle ownership, multi-functional collaboration, Agile

PROFESSIONAL EXPERIENCE

Sprouts AI | Software Engineer (Java / Server / Distributed) | Chicago, IL | Mar 2025 - Present

- Designed and built high-performance, scalable, and extensible server-side Java microservices in Core Java 17 and Spring Boot, sustaining 99.9% uptime for 1K+ daily users through asynchronous programming, non-blocking I/O, and multi-threading in low-latency distributed systems.
- Secured service-to-service communication with mutual TLS (mTLS), configuring certificate authorities, trust stores, certificate chains, hostname validation, and hardened TLS policies, and automating certificate rotation through secret management to prevent expiry-driven outages.
- Built Kafka-based event streaming for workflow lifecycle events with partitioning, offset management, and idempotency, resolving consumer-rebalance issues via lightweight poll loops, bounded worker pools, and commit-after-success semantics.
- Built security-auditing controls capturing authentication events, authorization decisions, certificate activity, and privileged operations as structured append-only records (actor, resource, action, trace ID), with SIEM alerts on repeated auth failures and abnormal access.
- Led the full lifecycle of multiple services from architecture and design to development, testing, and maintenance, shipping Docker images on Kubernetes (EKS) with workload identities and ingress gateways, CI/CD in AWS CodePipeline, and 80%+ JUnit 5/Mockito coverage.

Resilience Inc | Software Engineer (Java / Server / Security) | Chicago, IL | Aug 2023 - Mar 2025

- Built scalable server-side Java email and messaging services in Spring Boot on AWS handling 500K+ monthly requests, spanning message composition, recipient validation, template rendering, delivery orchestration, and status tracking with asynchronous programming and non-blocking I/O.
- Designed asynchronous delivery pipelines with message queues and event-driven services decoupling APIs from downstream email delivery, supporting transactional, scheduled, and bulk sends with retries (exponential backoff and jitter), idempotency, suppression lists, and dead-letter queues.
- Improved deliverability by implementing SPF, DKIM, DMARC, domain verification, bounce and complaint processing, provider failover, rate limiting, and delivery-state reconciliation, monitoring bounce rate, delivery latency, queue depth, and provider errors.
- Cut p95 read latency 35% with Redis caching over MongoDB/PostgreSQL (invalidation, TTL tuning, connection pooling) and reduced production incidents 45% via thoroughly tested code with JUnit 5 and Mockito wired into CI quality gates.
- Owned backend services end-to-end (architecture, design, development, testing, maintenance) with a privacy and security first approach, mTLS between microservices, and Docker/Kubernetes (EKS) Helm rolling releases that cut release cycles from 2 weeks to 2 days.

Tata Steel | Software Engineer (Java / Distributed / Cloud) | Apr 2018 - Aug 2023

- Architected enterprise server-side Java microservices in Core Java and Spring Boot on AWS and Azure with Docker and Kubernetes, achieving 99.99% uptime and 40% faster API responses across 5+ business domains serving 40K+ users.
- Designed Cassandra data models around access patterns with partition keys, clustering columns, denormalized tables, and time-bucketed partitions to avoid unbounded partitions and cross-node queries, applying CQL, replication strategies, consistency levels, compaction, and TTL-based retention.
- Operated Cassandra in production at scale, monitoring read/write latency, pending compactions, dropped mutations, tombstone scans, and node availability, resolving hot partitions and uneven token distribution, and tuning connection pools, retry policies, and idempotent writes with controlled schema migrations.
- Built Kafka event-driven workflows partitioned by employeeId + trainingId, resolving hot-partition issues through key design while preserving ordering, with multi-threading, asynchronous processing, and Redis caching for high availability under load.
- Led a 12-person engineering team through the full Agile SDLC with design reviews, JUnit/Mockito test strategy, and Jenkins CI/CD, cutting production incidents 50% with a strong sense of ownership from design to deployment.

PROJECTS

cassandra-mail-store - Distributed Mail Storage on Cassandra (Java, Cassandra, Redis) | [Source Code](#)

- Built a server-side Java mail storage service on Apache Cassandra, modeling mailboxes and messages with partition-key design for high-throughput reads/writes, replication, and tunable consistency, plus Redis caching, directly mirroring large-scale mail system architecture.

Real-time Stock Market Analysis Pipeline (Java, AWS, Kafka) | [Source Code](#)

- Built a high-performance streaming pipeline with server-side Java services, Kafka, AWS Lambda, REST APIs, and CloudWatch monitoring, applying asynchronous programming and non-blocking I/O for low-latency distributed processing.

PUBLICATIONS

IEEE - Comparative Study of Movie Recommendation Systems using Feature Engineering | [Paper](#) ([Source Code](#), [Certificate](#))

EDUCATION

Illinois Institute of Technology - M.S., Computer Science | National Institute of Technology, Calicut - B.S., Computer Science